DEBRE TABOR UNIVERSITY

GAFAT INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINERRING

COMPUTR ENGINEERING STREAM

TITLE: TOUR TICKET RESERVATION SYSTEM FOR GONDAR

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Submission Date: **Dec. 2025**
Debre Tabor, Ethiopia

Declaration

This semester project report entitled "Tour Ticket Reservation System for Gondar" has been submitted to the Department of Electrical and Computer Engineering in partial fulfillment of the requirements for the Semester Project in the Bachelor of Science in Computer Engineering program at Debre Tabor University. This project has been carried out by

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Abbreviations & Acronyms

Abbreviation	Full Meaning
API	Application Programming Interface
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
DB	Database
GPS	Global Positioning System
HTML	Hyper Text Markup Language
IDE	Integrated Development Environment
ISO	International Organization for Standardization
JS	JavaScript
MySQL	My Structured Query Language
PHP	Hypertext Preprocessor
SQL	Structured Query Language
TTRS	Tour Ticket Reservation System
UI	User Interface
UX	User Experience
WBS	Work Breakdown Structure

Abstract

The Tour Ticket Reservation System for Gondar is a web-based platform developed to modernize and automate the traditional manual ticketing process used at Gondar's historical and cultural attractions. The system addresses major challenges such as long queues, data inaccuracy, inefficient communication, and limited accessibility by providing tourists and administrators with a centralized digital solution. It enables tourists to browse attractions, register, book tickets, and make secure online payments, while administrators can manage attractions, ticket inventory, users, and generate real-time reports. The project was developed using Agile methodology, employing PHP, MySQL, HTML, CSS, and JavaScript to ensure flexibility, scalability, and user-friendliness. UML diagrams—including use case, activity, sequence, and object models—were used to support system analysis and design. The platform enhances operational efficiency, improves tourist satisfaction, promotes Gondar's heritage, and ensures reliable data management through role-based access control and security measures. Overall, the system provides a modern, efficient, and scalable digital solution that strengthens tourism management and supports economic growth in Gondar.

Acknowledgment

We would like to express our sincere gratitude to our advisor, Mr. Miftah, for his continuous guidance, support, and encouragement throughout the development of this Tour Ticket Reservation System project. His expert insights, constructive feedback, and dedicated supervision have been invaluable in shaping the direction and quality of our work. We deeply appreciate his patience, commitment, and timely advice, all of which have contributed significantly to the successful completion of this project.

We would also like to extend our heartfelt appreciation to our instructors, classmates, and family members for their unwavering support and motivation. Their assistance, encouragement, and belief in our abilities have played an essential role in helping us accomplish this project effectively.

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CHAPTER ONE

Introduction

Tourism is a vital sector that plays a significant role in the economic and social development of many countries. It contributes to GDP growth, job creation, infrastructure development, and cultural exchange. However, as the tourism industry expands, managing ticket reservations for various attractions and events becomes increasingly complex. This includes coordinating ticket sales, managing visitor information, handling bookings and payments, and maintaining effective communication among stakeholders.

A Tour Ticket Reservation System is a digital solution designed to automate and streamline the operations involved in managing ticket sales for tourist attractions and events. It provides a centralized platform for tourists, event managers, and administrators to interact efficiently. Through such a system, administrators can manage ticket inventory, schedule events, track sales activities, maintain venue information, and generate reports. Tourists, in turn, can browse attractions, book tickets, make payments, and receive confirmations—all from a singe platform.

1.1 Background of the project

Gondar, often referred to as the "Camelot of Africa," is a historic city in northern Ethiopia renowned for its rich cultural heritage and significant role in Ethiopian history. It served as the capital of the Ethiopian Empire from the 17th to 19th centuries and is located approximately 748 kilometers (465 miles) north of Addis Ababa. The driving distance between Addis Ababa and Gondar is about 748 kilometers, which typically takes around 12-14 hours by car, depending on road conditions and traffic.

Tourists visiting Gondar can explore several unique attractions that highlight its historical and architectural beauty. The Royal Enclosure (Fasil Ghebbi), a UNESCO World Heritage Site, is a fortress-city containing palaces, churches, monasteries, and other buildings from the 17th and 18th centuries. Fasiladas' Bath, another significant site, is used for the annual Timkat (Epiphany) celebrations, where thousands gather for the reenactment of baptism. Debre Berhan Selassie Church is famous for its magnificent ceiling decorated with winged angel faces and its well-preserved murals depicting biblical scenes.

Another notable attraction is the Castle of Fasiladas, the largest and most impressive castle in the Royal Enclosure, built by Emperor Fasiladas in the 17th century. Gondar is also known for its vibrant cultural festivals, particularly Timkat, which attracts both domestic and international tourists. The city's traditional music, dance, and handicrafts offer visitors a rich cultural experience.

During peak tourist seasons, such as major holidays or cultural festivals, tourists may experience waiting times that could span several hours for popular attractions like the Royal Enclosure or during Timkat celebrations. Waiting lines for ticket purchases and certain tourist sites may also be longer. To ensure a smoother experience, tourists are advised to plan ahead and book tickets and accommodations well in advance, particularly during high-traffic times like festivals or public holidays.

1.2 Statement of the problem

The overall activities in the existing ticket reservation system for Gondar's attractions are largely manual. The traditional system involves various ticket-related tasks such as ticket sales, visitor registration, venue information handling, reporting, and coordination between managers and ticket vendors.

However, this manual system has several challenges:

Inefficient and Error-Prone Operational Processes : The reliance on manual systems for core functions such as ticket scheduling, sales reporting, and visitor assignment is inherently time-consuming and prone to human error. This leads to frequent mistakes in data recording, processing delays, and unreliable information due to the high risk of data being lost, damaged, or mishandled.

Fragmented Communication and Ineffective Decision-Making: The absence of a unified platform creates critical breakdowns in coordination between managers, vendors, and tourists. This results in operational confusion, missed bookings, and poor service quality. Furthermore, the manual submission of critical data, like sales reports, creates significant delays, impeding timely analysis and strategic response by management.

Resource Inefficiency and Reduced Tourist Satisfaction: The manual framework necessitates unnecessary expenditure on physical materials (paper, printing, storage), increasing administrative costs. Simultaneously, it fails to provide tourists with the expected modern convenience of online booking and easy access to attraction information, directly undermining their experience and the service's overall appeal.

1.3. objective

1.3.1 General objective

The general objective of this project is to streamline tour ticket reservation management by developing a web-based system.

1.3.2. specific objective

Develop an integrated web-based platform that allows tourists to browse attractions, book tickets, make secure payments, and receive instant confirmations through a user-friendly, responsive interface.

Implement real-time communication and coordination tools for tourists, managers, and ticket vendors to ensure seamless interaction and efficient service delivery.

Automate core operational processes including ticket scheduling, visitor assignment, and reporting to reduce manual workload, improve efficiency, and eliminate paper-based procedures.

1.4. Scope of the project

The system incorporates a comprehensive set of core functionalities designed to enhance the tourism experience and streamline administrative operations. It includes robust user management features that allow tourists, event managers, and administrators to register and maintain their profiles. A dedicated tourism information portal provides detailed insights into Gondar’s historical landmarks, cultural attractions, and local experiences. Tourists can conveniently book tickets and make reservations in real time through the online ticketing module. Event managers can efficiently create and manage customized ticket packages, while tourists can share their experiences through the reviews and feedback system to support ongoing service improvement. System administrators retain full control over users, access privileges, and overall platform operations. The web-based platform ensures broad accessibility across desktops and laptops, supported by strong data security measures including secure login, password management, and privacy protections. Additionally, integration with payment services like cahpa ensures a seamless and efficient user experience throughout the platform.

1.5. Limitation of the project

The system faces several limitations that may affect its overall effectiveness. First, it is highly dependent on stable internet connectivity, which can be a challenge in remote or low-coverage areas, leading to reduced accessibility for some users. Additionally, the platform currently offers limited language support, which is only english, making it less accommodating for international tourists who may require multilingual options. Furthermore, the absence of a dedicated mobile application restricts accessibility to browser-based use only, which may limit convenience and usability for users who prefer mobile apps for seamless on-the-go access.

1.6. Significance of the project

The development and implementation of an Online Tour Ticket Reservation System for Gondar holds immense significance for local tourism authorities, attraction managers, tourists, and the broader economic development of the region. As Gondar continues to grow as a culturally and historically important destination, the digitization of its ticket reservation process provides transformative advantages. Below are the key aspects of its importance:

The system brings enhanced operational efficiency by digitizing key processes such as ticket sales, visitor management, venue bookings, and scheduling. This reduces manual workload, minimizes human errors, and streamlines daily tasks for tourism offices and service providers. Tourists benefit from a significantly improved experience, gaining access to accurate and up-to-date information on attractions and events. They can plan their visits, book tickets, and receive real-time notifications, all of which improve convenience, safety, and overall satisfaction.

Beyond operational improvements, the platform plays an essential role in promoting Gondar's cultural heritage. Through features like virtual galleries, interactive maps, historical descriptions, and multimedia content, the system showcases the city's rich identity to both local and global audiences. This strengthens cultural preservation efforts while increasing tourist interest. At the same time, the web-based nature of the platform expands access to tourism services—enabling rural residents and international visitors to book tickets remotely and discover local attractions. This improves inclusivity, supports local businesses, and broadens community involvement in the tourism economy.

The system also contributes to long-term economic growth and sustainability. By cutting paperwork, reducing administrative overhead, and enabling self-service features, it lowers operational costs for tourism institutions and saves time and resources for visitors. Its built-in analytics and reporting tools empower stakeholders with real-time insights that support data-driven decision-making, strategic planning, and effective marketing. With a scalable architecture that supports future enhancements such as QR code ticketing, advanced payment gateways, and expanded feedback systems the platform positions Gondar as a competitive and modern tourist destination, driving increased visitor flow and strengthening the region's economic development.

CHAPTER TWO

Literature Review

Tourism in Gondar, a UNESCO World Heritage site, is a vital component of its economic and cultural landscape. This literature review explored the research conducted on tourism systems in Gondar, focusing on key areas such as tourism development, infrastructure, cultural preservation, economic impact, tourist satisfaction, and sustainable practices. A variety of research methods were employed in these studies, providing insights into the effectiveness and challenges of tourism management in the city.

UNESCO World Heritage Centre [1] emphasized that the Fasil Ghebbi (Royal Enclosure) in Gondar was of outstanding universal value and required a comprehensive management plan that balanced heritage conservation with sustainable tourism development.

Gashaw and Alemu [2] identified critical issues in Gondar's heritage management, including visitor congestion, poor coordination among tourism stakeholders, and the absence of systematic tourist data management. Their findings highlighted the need for a Tour Ticket Reservation System (TTRS) to address these challenges.

Bekele and Yilma [3] analyzed the role of ICT in Ethiopian tourism and concluded that digital platforms could enhance promotion and accessibility; however, they highlighted infrastructure limitations, cost, and low digital literacy as major barriers to implementing systems like the TTRS.

Tadesse [4] discussed broader obstacles to digital transformation in Ethiopia's tourism industry, stressing the need for stakeholder training and institutional support for technology adoption, which informed the training components of the TTRS implementation.

Fenta and Tilahun [5] examined the impact of tour-guide service quality on visitor satisfaction at Gondar heritage sites, finding that guide professionalism and communication directly influenced tourist experience and site preservation. This research supported the user experience design principles incorporated into the TTRS.

Fikadu [6] analyzed the legal framework governing digital payments in Ethiopia and confirmed that current legislation supported e-commerce and online financial transactions, which enabled the secure payment processing features of the TTRS.

Gupta and Kumar [7] demonstrated that integrating ICT tools such as GIS mapping, CRM systems, and analytics dashboards improved tourism planning, crowd management, and heritage protection. Their research informed the technical architecture of the TTRS.

Mengistu and Kassa [9] presented empirical evidence that applying GIS technology in Ethiopian tourism enhanced sustainable resource management and supported data-driven decision-making, which influenced the reporting and analytics features of the TTRS.

Therefore, existing literature underscored the importance of ICT-based solutions for efficient heritage management but revealed a gap in locally tailored, integrated TMS platforms specifically designed for Gondar's unique context. This gap was addressed by developing the Tour Ticket Reservation System (TTRS) described in this documentation.

CHAPTER THREE

Methodology

3.1. System development methodology

Agile is the most suitable methodology for developing the Tour Ticket Reservation System because it supports gradual system development while ensuring continuous collaboration among developers and stakeholders. Its flexibility allows necessary modifications to be incorporated even in the later stages. Through constant communication between developers, tourists, event managers, and administrators, Agile ensures that real-world requirements are quickly understood and integrated into the system. The iterative testing process further strengthens the project by enabling continuous evaluation of system features, allowing immediate identification and correction of issues. This approach improves the reliability, usability, and overall functionality of the final reservation system, ensuring it effectively meets the needs of tourists and supports efficient management for Gondar's tourism services.

3.2 Data collection methodology

3.2.1. Document and Literature Review

The research will focus on identifying best practices in ticket reservation management, analyzing current trends, and exploring technology solutions that can be integrated into the system. By conducting a thorough review of relevant literature and case studies from other ticketing systems, the study will ensure that the proposed system incorporates the most up-to-date and effective strategies. This approach will help in adopting proven methods and innovative technologies to enhance the efficiency and overall experience of tour ticket reservation management.

3.2. Hardware and software requierment

Table 3.1. Hardware tools

Hardware	Specification	Purpose
Desktop Computer/PC	Core i7	Server an storage
Flash Disk 8GB	16GB	Store and transfer

Table 3.2 software tool table

Tool/Technology	Activities
HTML	Designing the structure and layout of web pages for the (TTRS).
CSS	Styling and formatting web pages to enhance visual appeal and user experience.
JavaScript	Adding interactivity, such as dynamic content updates and form validation.
PHP	Developing server-side logic for user authentication, bookings, and data processing.
XAMPP	Setting up a local testing environment to run the system before deployment. Like MySQL and Apache.
Visual studio code	For editing code

3.4. System Requirements

System requirements refer to the minimum set of hardware, software, and network resources needed for a computer system, application, or platform to function correctly. These requirements ensure that the system operates smoothly, reliably, and securely. They typically include specifications for the operating system, memory (RAM), storage space, processor type, software dependencies, and sometimes internet connectivity or browser compatibility.

3.4.1 Existing System Overview

The current ticket reservation system for Gondar's attractions is largely traditional and manual, with limited use of digital tools or centralized platforms. Most activities, including ticket sales, visitor coordination, and attraction information services, are handled in person or through informal communication. There is no unified online system where tourists can plan visits, book tickets, or learn about Gondar's cultural and historical attractions in real time. Marketing efforts are minimal and not well-organized on digital platforms, which reduces Gondar's visibility to both domestic and international tourists. In addition, collaboration among tourism offices, attraction managers, and cultural institutions is unstructured, leading to inefficiencies and inconsistent service delivery. Due to the absence of data collection and reporting systems, tourism authorities lack accurate insights into visitor trends, service quality, and growth opportunities.

3.4.2 Drawback of the Existing system

The existing tour management system faces several inefficiencies that affect both tourists and service providers. Reliance on traditional, paper-based processes leads to delays, errors, and difficulties in managing ticket sales efficiently. Tourists often struggle to access reliable, up-to-date information about attractions, schedules, and pricing. Furthermore, the absence of an online ticket booking and reservation system prevents visitors from planning and securing tickets in advance. Communication between tourists and service providers is also weak, as there is no centralized platform for direct interaction with attraction managers or ticket vendors.

3.5. Proposed System of the project

The Online Tour Ticket Reservation System for Gondar is a comprehensive digital platform designed to modernize and streamline ticket reservation operations in the region. This system will facilitate smooth interactions between tourists, attraction managers, ticket vendors, and local authorities, enabling better management, coordination, and promotion of Gondar's rich cultural and historical offerings.

Currently, ticket reservation operations in Gondar are managed manually, causing inefficiencies and a lack of coordination. The proposed system will address these challenges by providing a centralized platform for all stakeholders involved in the tourism sector.

Key components of the proposed system include:

- **Tourist Portal**: A user-friendly website where tourists can explore Gondar's attractions, book tickets, and receive information about events and cultural sites. This portal will provide an interactive booking, detailed descriptions, and booking options for each tourist spot.
- **Payment Gateway Integration**: The system will support secure online payment gateways for tourists to pay for tickets and services, making the transaction process seamless and efficient (like chapa).

3.5.1.Functional Requirement

Here are the functional requirements for the Tour Ticket Reservation System for Gondar summarized:

System Features: The tour ticket booking platform is designed to offer a seamless and enjoyable experience for both tourists and service providers. **User Management** allows effortless registration, secure login, profile updates, and role-based access control. **Ticket Package Management** makes it easy to add, update, delete, search, categorize, and view ticket details at a glance. The Booking System ensures smooth online reservations, real-time availability checks, and hassle-free modifications. With the **Payment System**, users enjoy secure transactions through Chapa, instant invoices, and easy refund processing. Finally, **Reviews & Feedback** empowers users to share their experiences, while admins can monitor and moderate to maintain quality and trust.

3.5.2. Non-Functional Requirement

Non-functional requirements define how the Tour Ticket Reservation System (TTRS) should perform beyond its core functionalities. These requirements focus on usability, reliability, performance, supportability, and legal compliance to ensure a high-quality user experience and system effectiveness.

Usability The system should be user-friendly, intuitive, and easy to navigate for all user types—tourists, event managers, and administrators. It should minimize the time and effort required to complete tasks like booking tickets, managing schedules, or generating reports.

Reliability The TTRS must operate consistently and accurately, with minimal downtime. It should maintain data integrity, provide real-time notifications in case of failures (e.g., booking errors or payment failures), and support regular backups to prevent data loss.

Performance The system must be fast and responsive, capable of efficiently handling a growing number of users, bookings, and data. Page loads, search operations, and data transactions should occur within a few seconds, even during peak tourism seasons.

Supportability The system should offer technical documentation, user manuals, and access to a help desk or support center. It must also be designed for easy maintenance and debugging, ensuring smooth updates and long-term sustainability.

Legal The TTRS must comply with national and regional laws governing tourism, data protection, and online transactions (e.g., GDPR or Ethiopia's ICT policy). It should ensure user consent for data collection and include features for audit trails and periodic compliance checks.

3.6. Requirements Analysis

3.6.1 System models

A system model is a conceptual representation of a system that defines its structure, behavior, and interactions. It provides a simplified view of complex real-world processes, helping to analyze, design, and optimize systems efficiently. System models can take various forms, such as diagrams, mathematical models, or simulations, depending on the domain.

3.6.1.1 Use case model

A Use Case Model is a visual representation of a system's functionality from a user's perspective. It consists of actors (users or external systems), use cases (specific interactions), and relationships between them. This model helps in understanding system requirements by depicting how users interact with the system to achieve their goals. It is widely used in software development to define functional requirements and ensure that the system meets user needs efficiently. The Unified Modeling Language (UML) is commonly used to create use case diagrams, making them a crucial tool in requirement analysis and system design. The system has the following actors:

1. Tourist
2. System Admin

Tourist can

Register Login View Available Attractions Book Ticket Make Payments Contact Support Provide Feedback

System Admin can

Login Manage Attractions Manage Users and packages View Booking Statistics System Configuration

3.6.1.2 Use case diagram

Figure 3.1. usecase diagram

Use Case Descriptions table

Table 3.3 Description for Sign Up

Use Case ID	UC-1
Use Case Name	Sign Up
Actor(s)	Tourist
Description	Tourist creates system account
Precondition	No existing account
Basic Flow	1. Open registration page 2. Fill details 3. Submit form 4. Verify email 5. Account created
Post Condition	Account active

Table 3.4 Description for Sign In

Use Case ID	UC-2
Use Case Name	Sign In
Actor(s)	Tourist, System Admin
Description	User logs into system
Precondition	Account exists
Basic Flow	1. Enter credentials 2. System validates 3. Access granted
Post Condition	User logged in

Table 3.5 Description for View Available Tickets

Use Case ID	UC-3
Use Case Name	View Available Tickets
Actor(s)	Tourist
Description	Tourist views ticket availability
Precondition	None
Basic Flow	1. Navigate to tickets 2. View list 3. Check availability
Post Condition	Ticket options displayed

Table 3.6 Description for Book Tickets

Use Case ID	UC-4
Use Case Name	Book Tickets
Actor(s)	Tourist
Description	Tourist reserves tickets
Precondition	Tickets available
Basic Flow	1. Select tickets 2. Choose quantity 3. Reserve 4. Proceed to payment
Post Condition	Tickets reserved

Table 3.7 Description for Make Payment

Use Case ID	UC-5
Use Case Name	Make Payment
Actor(s)	Tourist
Description	Tourist pays for tickets
Precondition	Tickets reserved

Basic Flow	1. Select payment method 2. Enter details 3. Process payment 4. Confirm
Post Condition	Payment completed

Table 3.9 Description for Manage Ticket Inventory

Use Case ID	UC-6
Use Case Name	Manage Ticket Inventory
Actor(s)	System Admin
Description	Admin manages ticket stock
Precondition	Admin logged in
Basic Flow	1. Access inventory 2. Update quantities 3. Save changes
Post Condition	Inventory updated

Table 3.10 Description for Generate Reports

Use Case ID	UC-7
Use Case Name	Generate Reports
Actor(s)	System Admin
Description	Admin creates sales reports
Precondition	Admin logged in
Basic Flow	1. Select report type 2. Set parameters 3. Generate report 4. View/download
Post Condition	Report generated

Table 3.11 Description for Cancel Booking

Use Case ID	**UC-8**
Use Case Name	**Cancel Booking**
Actor(s)	**Tourist**
Description	**Tourist cancels booking**
Precondition	**Booking exists**
Basic Flow	**1. View bookings** **2. Select booking** **3. Cancel** **4. Confirm cancellation**
Post Condition	**Booking cancelled**

3.6.1.3. Block Diagram

A Block Diagram is a high-level graphical representation of a system that illustrates the main components or modules and the flow of information between them, helping to simplify and visualize how the entire system operates as an integrated whole.

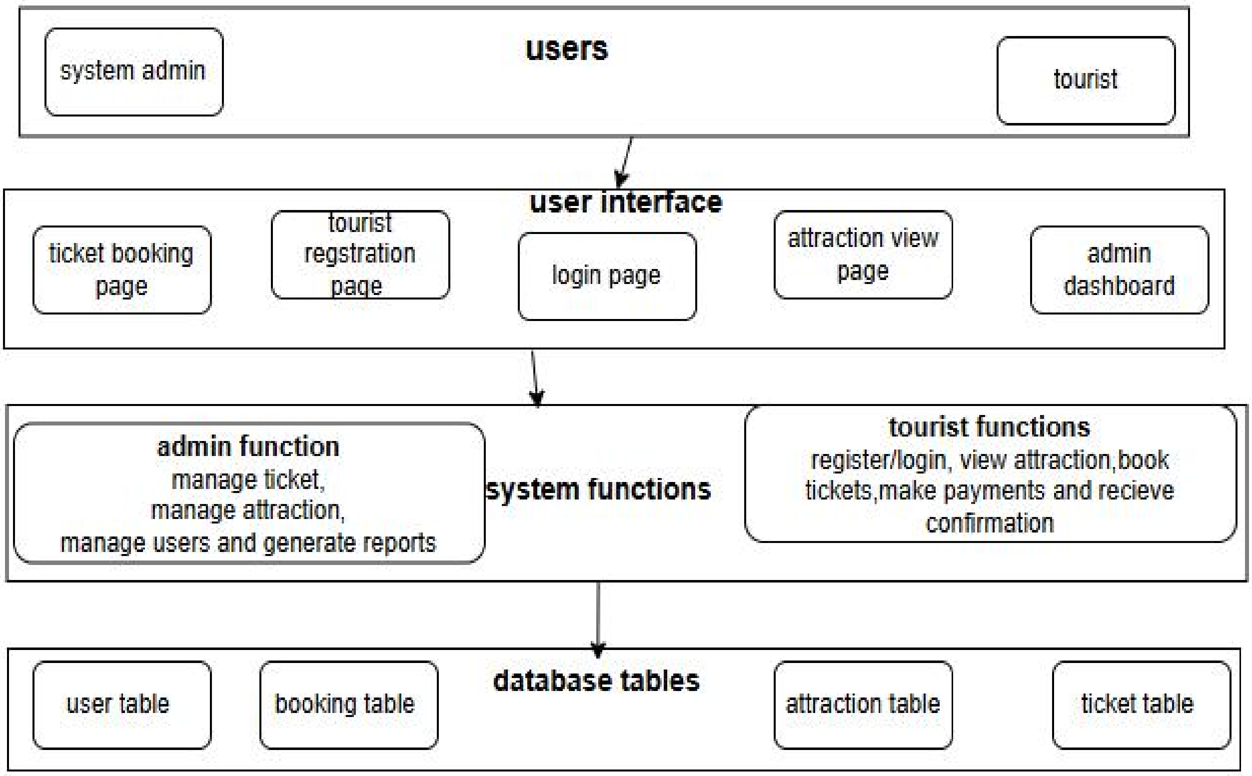

Figure 3.2 Object Diagram

3.6.1.4. Sequence Diagram

A **sequence diagram** is a UML diagram that shows the interaction between system components over time through message exchanges. It helps visualize the flow of a process, making it useful for system design and documentation.

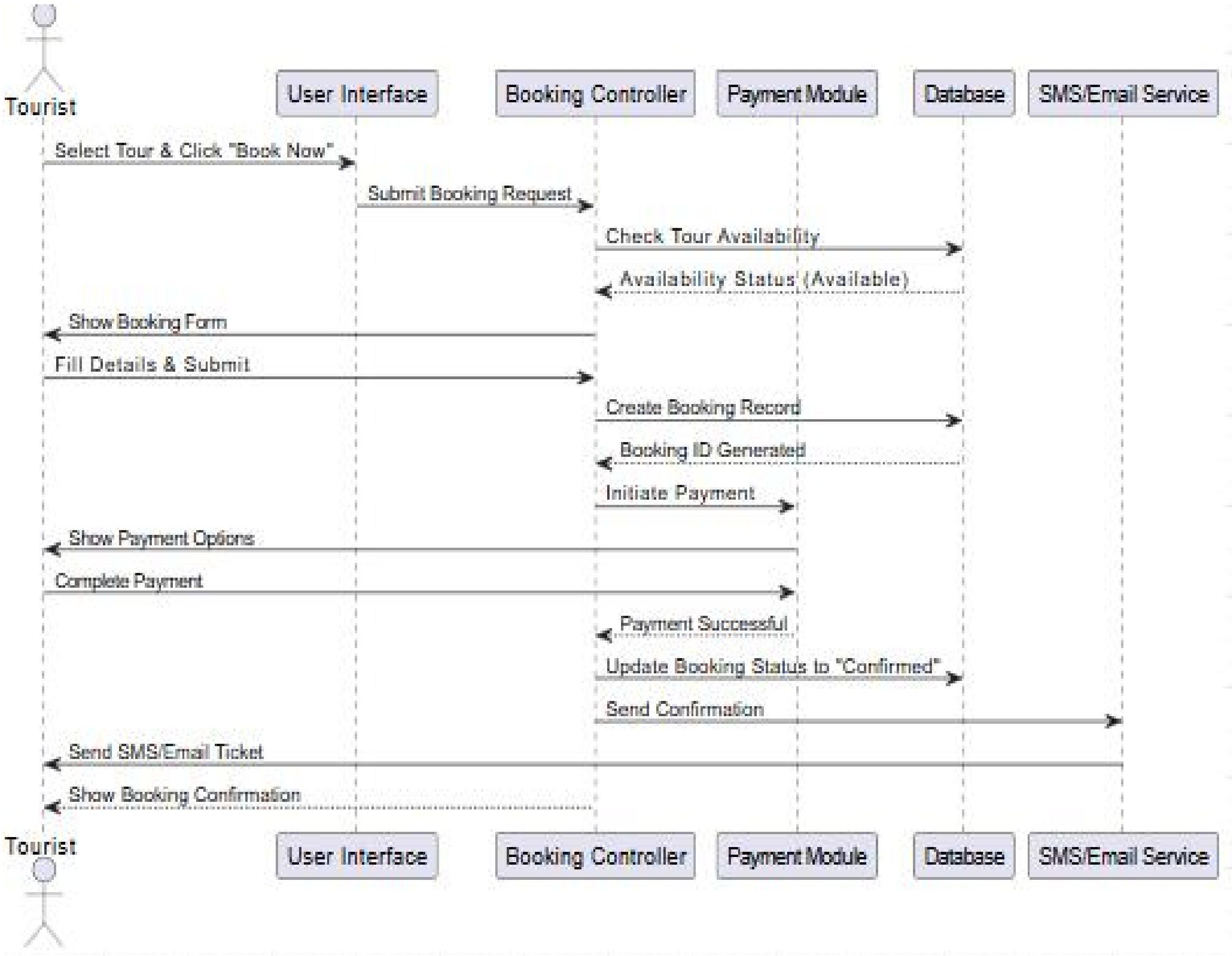

Figure 3.3 Sequence Diagram for Ticket Booking

Sequence Diagram for Admin management

In an admin management sequence diagram, the administrator sends a request through the system interface, and the system checks if the admin is authorized. After verification, the backend processes the request and interacts with the database to update or retrieve information. The system then returns a response to the admin, confirming the result of the operation.

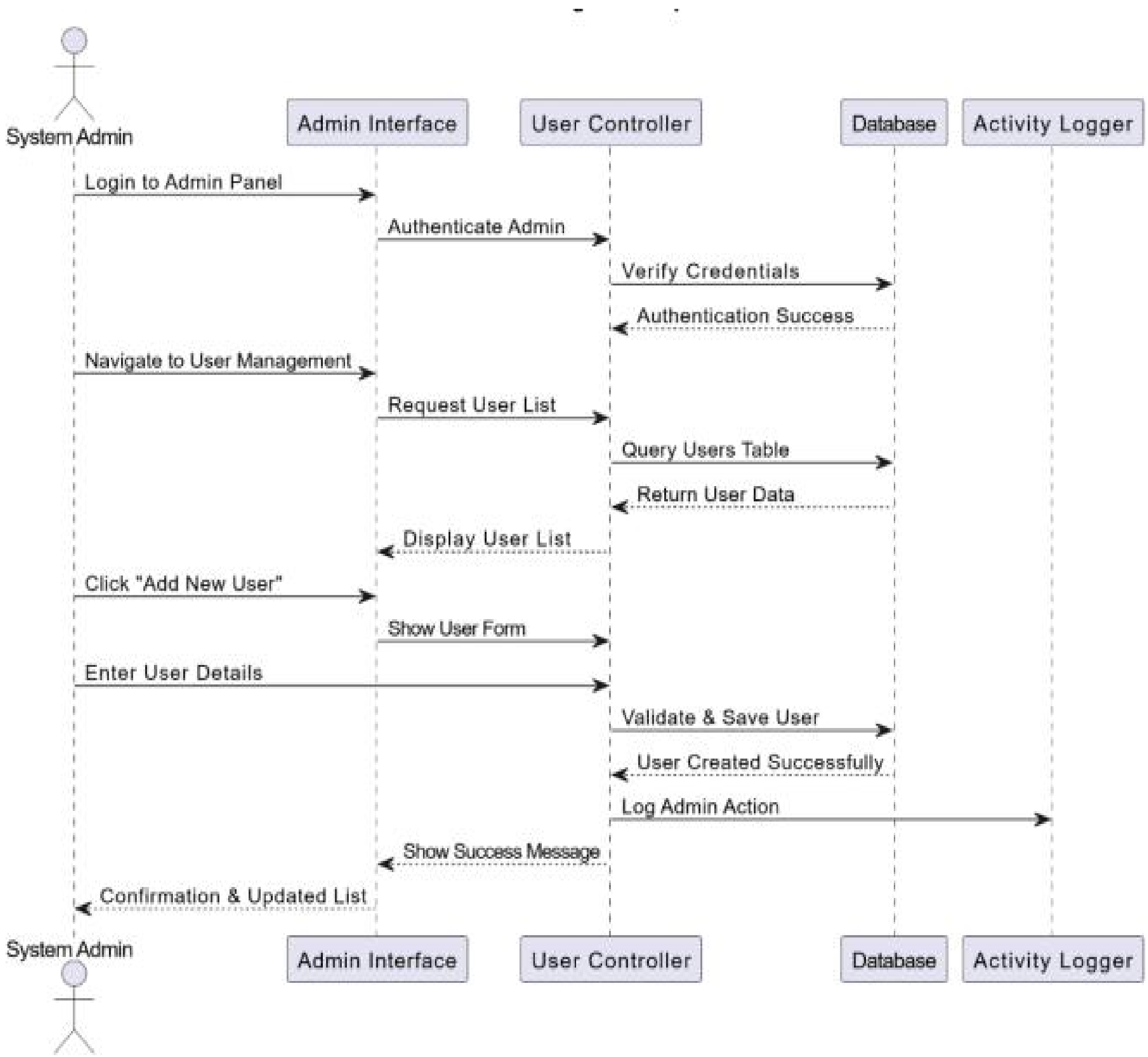

Figure 3.4 Sequence Diagram for Admin management

3.6.1.5. Activity diagram

An activity diagram is a visual representation of the workflow or process within a system, showing the sequence of activities, decisions, and actions that occur. It highlights how users or systems interact, the flow of control between different actions, and the conditions under which certain paths are followed. Activity diagrams are commonly used in software design to model business processes, user interactions, and system workflows.

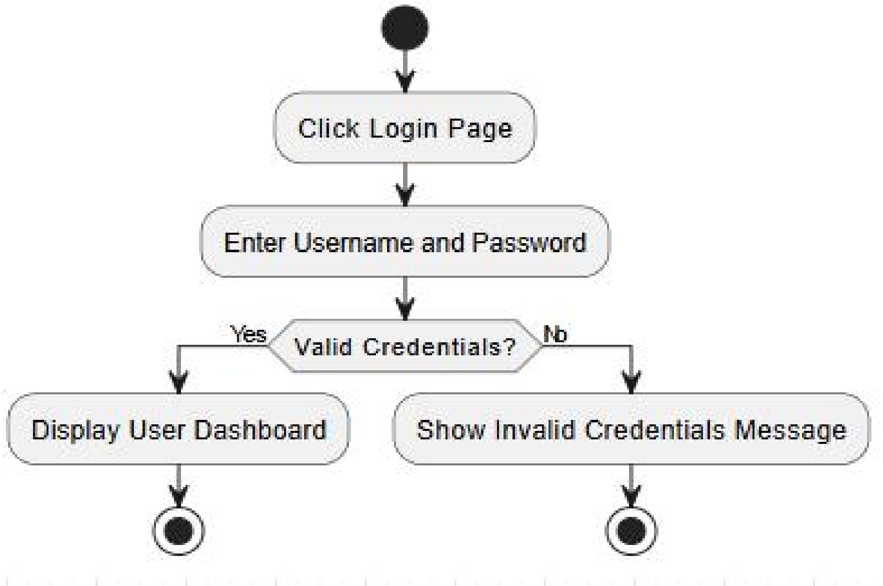

Figure 3.5 activity diagram for login

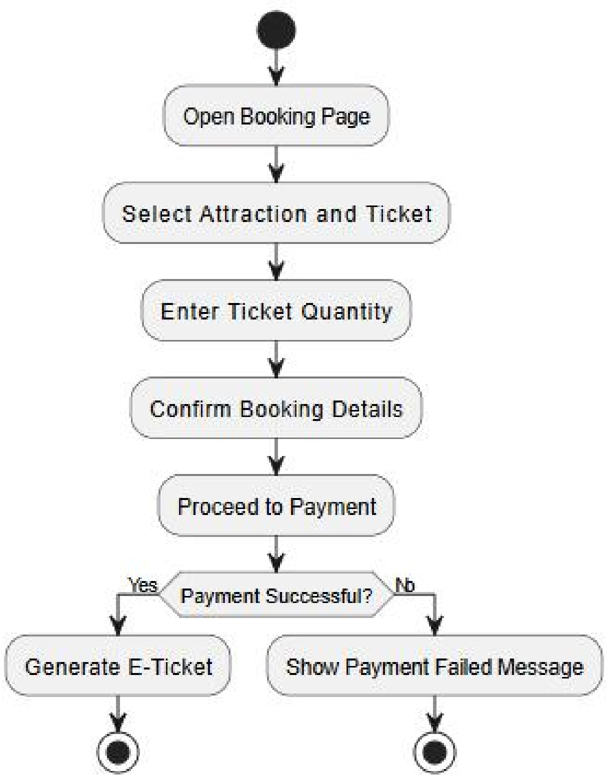

Figure 3.6 activity diagram for booking tickets

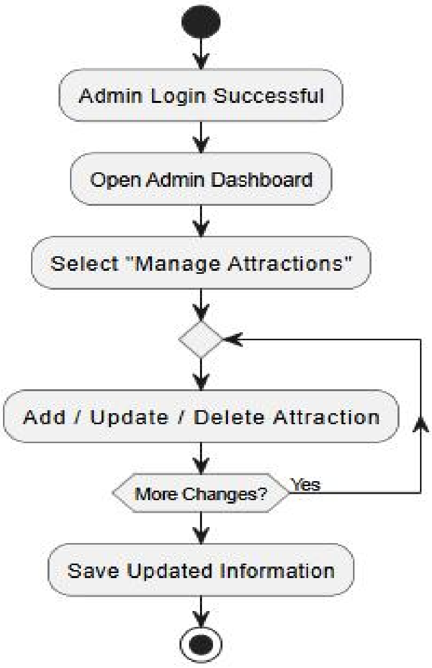

Figure 3.5 Activity diagram for admin management

3.7. Feasibility Analysis

The tour ticket registration system is economically feasible because it reduces the cost of manual ticket handling, cuts paper-related expenses, and improves revenue accuracy through automated sales tracking. Its ability to attract more tourists through convenient online access increases overall income, making the system financially beneficial in both the short and long term.

Politically, the above system is feasible because it aligns with government initiatives promoting digital transformation and improved public service delivery. It supports national tourism strategies, enhances transparency, and encourages more organized management of cultural sites, which makes it acceptable and supported by local authorities.

Socially**,** the above system is feasible, it improves tourist convenience, reduces long queues, and provides reliable access to attraction information. It also supports cultural awareness by showcasing Gondar's heritage digitally, which benefits both local communities and visitors. The system enhances user satisfaction and promotes inclusive access for a wider range of tourists.

CHAPTER FOUR

System design, development and implmentation of online tour ticket reservation system

Introduction

System design is the process of planning the architecture and components of a software system. It translates requirements into a blueprint, detailing how the system will function, how components will interact, and how data will flow. Key elements include defining the system architecture, selecting technologies, and ensuring scalability, security, and usability. The design serves as a guide for development, ensuring the system meets both functional and non-functional requirements efficiently.

4.1. System Design

The system design of the Tour Ticket Reservation System for Gondar focuses on building a structured, efficient, and user-friendly platform that meets both the functional and non-functional requirements. The design phase outlines how the system components interact to deliver core services such as attraction browsing, ticket booking, payment processing, and administrative management.

The architecture is designed to support modularity, scalability, and maintainability, ensuring that each system function can be developed and updated independently. The user interface is planned to be intuitive and accessible, catering to different user roles: tourists and system administrators. Security mechanisms such as login authentication and role-based access control are integrated to protect sensitive data and ensure only authorized users can access specific functionalities.

The system also emphasizes high performance, aiming for fast response times and real-time data updates. By combining clear navigation, logical workflow, and strong backend support using PHP and MySQL, the design ensures that the system will provide reliable and efficient ticket reservation services tailored to the needs of Gondar's tourism attractions.

4.1.1. Design Goals

The primary objective of the system design phase is to model the Tour Ticket Reservation System for Gondar with high quality and clarity, ensuring it meets both functional and non-functional requirements. In this context, non-functional requirements describe the system's attributes, such as performance, security, usability, and any constraints that define the boundaries of the solution.

These non-functional requirements directly influence the design goals, which guide the development team in achieving a system that is not only functional but also robust, reliable, and user-friendly. The system should minimize time, cost, and effort in managing ticket reservation tasks such as attraction browsing, ticket booking, payment processing, and administrative oversight. It must deliver accurate results with minimal resource waste, ensuring quick processing and effective task completion. To protect sensitive data, the system must require user authentication through valid credentials. Only authorized users (tourists and system administrators) can access the system's features according to their role, ensuring data privacy and system integrity. Payment transactions must be encrypted and secure. The system should exhibit high responsiveness, even under load. Actions such as viewing attractions, booking tickets, or generating reports should be executed swiftly, with minimal lag, to support real-time interactions and prevent user frustration during peak booking periods. The system interface should be simple, intuitive, and accessible. Forms, buttons, and pages should be clearly labeled with descriptive names. Whether used by tourists or system administrators, the system must reduce the learning curve and make navigation effortless, ensuring a positive user experience. The system must operate consistently with minimal downtime. Data integrity must be maintained, and backup mechanisms should be in place to prevent data loss. The system should recover gracefully from failures and provide clear error messages when issues occur. Scalability: The design should accommodate growth in user numbers, attraction listings, and transaction volumes without significant re-engineering. The architecture should support horizontal scaling when needed.

4.1.2. Current Software Architecture

The existing ticket reservation system for Gondar's attractions is a manual system, and there is no current software architecture in place. The system relies on paper-based record-keeping, physical ticket sales at attraction entrances, and in-person interactions. There is currently no integrated digital infrastructure for ticket reservations, leading to inefficiencies, data redundancy, and limited accessibility for tourists. Efforts are underway to transition to a modern digital system to enhance service delivery and improve the visitor experience.

4.1.3. Proposed Software Architecture

The Proposed Software Architecture for the Online Tour Ticket Reservation System of Gondar is designed to address the limitations of existing manual ticketing systems, enhance user experience, and scale efficiently to meet increasing demand. The architecture ensures high performance, flexibility, and maintainability while addressing the specific operational needs of tourists and system administrators.

To modernize ticket reservation services in Gondar, we propose a centralized, web-based, and modular software architecture that integrates essential functionalities such as tourist registration, attraction browsing, ticket booking, payment processing, and administrative management.

Authentication Module: Handles user registration, login, session management, and password recovery.

Attraction Management Module: Manages attraction data, including descriptions, images, prices, and availability.

Booking Engine: Processes ticket reservations, checks availability, and manages booking lifecycle.

Payment Processing Module: Integrates with payment gateways for secure transaction handling.

Reporting Module: Generates sales reports, visitor statistics, and system analytics.

Notification Service: Sends email and SMS confirmations, reminders, and alerts.

Search and Filter Module: Enables efficient attraction discovery based on various criteria.

4.1.4. Hardware/Software Mapping

One of the major tasks in system design, hardware/software mapping defines the allocation of components to specific hardware. The proposed Tour Ticket Reservation System for Gondar uses personal computers for administrators and servers for hosting the web application and database. Cloud storage and external drives are used for backups and system files.

The system runs on Microsoft Windows 10/11, with the front-end developed using HTML, CSS, JavaScript, and PHP. The back-end uses XAMPP and MySQL for server setup and data storage. Microsoft Office 2021 is used for documentation. This hardware/software mapping ensures efficient operation and scalability for the system.

Table 4.1: Hardware Specification

No.	List of Tools	Specification	Function
1	**Personal Computer/desktop**	CPU: Intel(R) Core i7 RAM: 8GB Storage: 512GB SSD	Used by developers and administrators for coding, testing, and accessing the system.
2	**Server**	RAM: 32GB	Hosts the application backend, databases, and handles data processing and storage.
4	**Flash Drive**	SanDisk 64GB USB 3.0	Used for transferring data and system backups between devices.

Table 4.2: Software Specification

No.	List of Tools	Specification	Function
1	**Operating System**	Windows 10 cori 7	For development, deployment, and testing of the system.
2	**Database**	MySQL,Apache, Xampp	To store and manage structured and unstructured

			data for bookings, users, and tourism information.
3	**Version Control**	Git, GitHub	To manage source code, track changes, and enable collaboration during system development.
4	**IDE (Integrated Development Environment)**	Visual Studio Code	For coding, testing, and debugging the system's software components.

4.1.5. Database Design

A relational database was designed using MySQL to store and manage all ticket reservation data. The schema was normalized to reduce redundancy and ensure data integrity. Key tables included:

The system manages **users**, storing login credentials and roles such as admin and tourist, and **tourists**, which includes profile data like name and contact information, linked to the user accounts. It also handles **tickets**, including types, prices, and availability, as well as **bookings**, which track ticket reservations and their status, and **payments**, recording both completed and pending transactions.

4.1.6. Data Dictionary

A data dictionary provides a detailed description of the data used in the system, including field names, data types, sizes, and their purpose. Below is the data dictionary for the Tour Ticket Reservation System.

Table 4.3: Dictionary for User Table

Field Name	Data Type	Size	Description
user_id	INTEGER	11	Primary key, auto-increment
first_name	VARCHAR	50	User's first name
last_name	VARCHAR	50	User's last name
email	VARCHAR	100	Unique email address

password	VARCHAR	255	Hashed password
phone	VARCHAR	20	Contact number
Created at	DATETIME	-	Account creation timestamp

Table 4.4: Dictionary for Booking Table

Field Name	Data Type	Size	Description
Booking id	INTEGER	11	Primary key, auto-increment
User id	INTEGER	11	Foreign key to users table
Visit date	DATE	-	Scheduled visit date
Ticket qty	INTEGER	11	Number of tickets booked
Total amount	DECIMAL	10,2	Total booking amount
Booking status	VARCHAR	20	Status (confirmed, cancelled, etc.)
Booking date	DATETIME	-	Booking creation timestamp
Confirmation code	VARCHAR	50	Unique confirmation identifier

Table 4.5: Dictionary for Payment Table

Field Name	Data Type	Size	Description
Payment id	INTEGER	11	Primary key, auto-increment
Booking id	INTEGER	11	Foreign key to bookings table
amount	DECIMAL	10,2	Payment amount

Payment method	VARCHAR	50	Method used (card, mobile, etc.)
Transaction id	VARCHAR	100	Gateway transaction ID
Payment status	VARCHAR	20	Status (completed, failed, etc.)
Payment date	DATETIME	-	Payment timestamp
Receipt url	VARCHAR	255	Receipt download URL

Table 4.6: Dictionary for Ticket Table

Field Name	Data Type	Size	Description
Ticket id	INTEGER	11	Primary key, auto-increment
date	DATE	-	Valid visit date
Available qty	INTEGER	11	Available tickets count
price	DECIMAL	10,2	Current price
status	VARCHAR	20	Status (available, sold_out, etc.)

4.1.7. Access Control and Security

Access control and security define what actions users are authorized to perform within the system. Implementing effective access controls is critical to protect the system's data integrity, prevent unauthorized access, and ensure that each user interacts only with information relevant to their role. The Tour Ticket Reservation System uses a role-based access control (RBAC) mechanism combined with technical and administrative security measures to safeguard sensitive data such as user profiles, ticket bookings, and financial transactions.

The following table outlines access control rules for the primary system roles: System Admin and Tourist.

Table 4.7: Access Control

Role	Functions	View	Modify	Create	Delete
Admin	Manage users Access all data Generate reports configure system Approve tours revie reports update policies	All System Data, All Tour & User Data	All Data, Tour Assignments,	All Accounts & Content, New Tour Packages, Guide Assignments	All Accounts & Content, Tour Packages,
Tourist	Book tour view details update profile make payment	Own Bookings, Tour Details	Personal Profile	Booking Requests, Reviews	Cancel Bookings (Before Due)

4.1.8. Boundary Conditions

Boundary conditions define the external constraints and internal dependencies that limit or influence the system's operation. For the Tour Ticket Reservation System for Gondar, the system is designed to support two key actors: System Administrator and Tourist. The following boundary conditions are outlined for initialization, termination, and failure recovery.

Table 4.8 Boundary Conditions (Initialization)

Name	Start Tour Ticket Reservation System
Actors	- System Administrator (manual startup) - Tourist (automatic upon availability)
Entry Conditions	- System servers inactive (for Admin) - Platform online and accessible (for Tourist)
Flow of Events	- **Admin**: Manually verifies and initializes core system services. - **Tourist**: Upon access, system verifies account and provides access to ticket browsing and booking features.
Exit Conditions	- System becomes fully operational.

	- Each actor receives appropriate interface ready for interaction.
Exceptions	- Admin receives error logs if startup fails. - All users receive maintenance or system error notifications if access fails.
Special Requirements	- Admin must have high-level access credentials. - All users require valid accounts and internet connectivity.

Table 4.9 Boundary Conditions (Termination)

Name	Shut Down Tour Ticket Reservation System
Actors	- System Administrator (controlled shutdown) - System (automatic on failure)
Entry Conditions	- Platform actively serving users (controlled) - Critical error or hardware/software failure (automatic)
Flow of Events	**Controlled Shutdown**: • Admin initiates shutdown via interface or command. • System halts user activity and saves session data. • System status set to "Offline". **Automatic Shutdown**: • System attempts self-repair. • If unsuccessful, fails over to backup or terminates. • Admin is alerted with logs and system status.
Exit Conditions	- All services terminated; data secured. - All actors notified of shutdown or maintenance.
Special Requirements	- Admin must authenticate shutdown rights.

Table 4.10 Boundary Conditions (Failure Recovery)

Name	Handle System Failure
Actors	- System (automatic recovery) - System Administrator (manual intervention)
Entry Conditions	- Unexpected hardware/software failure detected.
Flow of Events	- System triggers predefined self-recovery actions (restart services, rollback). - If recovery fails, switches to backup servers. - Admin investigates logs, performs diagnostics, and manually restarts or initiates maintenance.
Exit Conditions	- System resumes operation or transitions into controlled maintenance mode.
Special Requirements	- Redundant infrastructure and robust recovery mechanisms. - Admin must have access to real-time system monitoring tools and logs.

Explanation of Terms

Initialization (Startup Use Cases):

The process of transitioning the system from an offline or idle state to an operational state. Involves verifying services, loading role-specific dashboards, and confirming user access.

Termination (Shutdown Use Cases):

Describes how the system is gracefully shut down by the Admin or automatically due to internal issues, ensuring data integrity and user notifications.

Failure Recovery (Failure Use Cases):

Encompasses automatic and manual recovery procedures following unexpected system issues, aiming to minimize downtime and data loss.

4.2. Development of Tour ticket reservation system

This section provides an overview of the development process of the Online Tour Ticket Reservation System, a web-based platform designed to facilitate ticket reservation operations in Gondar by connecting tourists and system administrators. The primary goal of the system is to streamline the process of ticket browsing, booking, payment processing, and reservation management through a centralized digital solution.

The development followed a modular and iterative approach, ensuring that each component was developed, tested, and integrated effectively. The system architecture was designed to support scalability, reliability, and ease of use across both user roles. Key technologies used in the development include PHP for backend logic, MySQL for database management, HTML/CSS and Bootstrap for responsive front-end design, and JavaScript for client-side interactivity.

The development process was divided into several phases, including requirements gathering, database and UI design, coding of system modules, integration of payment gateways, and testing. Emphasis was placed on usability, performance, data security, and role-based access control to meet the needs of diverse users. The final product ensures real-time ticket availability checking, secure payment processing, easy management of reservation data, and a user-friendly interface accessible on multiple devices.

4.2.1 Data Preparation / Hardware Preparation

Hardware Preparation

Before the development began, both data and hardware requirements were assessed and prepared as follows:

Hardware Setup:

- **Development environment**: Intel Core i5, 8GB RAM, 256GB SSD.
- **Server setup**: XAMPP (Apache, PHP, MySQL) for local testing.
- **Smartphones and tablets** used for responsive interface testing.

Software Tools:

- **Backend**: PHP (with MySQL database)
- **Frontend**: HTML, CSS, JavaScript, Bootstrap
- **Code Editor**: Visual Studio Code

Data Preparation:

Data preparation is a crucial step in the development of the Tour Ticket Reservation System. It involves organizing and structuring the necessary data required for both system implementation and testing. This step ensures that realistic, consistent, and secure datasets are used throughout the development process to validate the system's functionality and performance.

1. Media and Asset Preparation

Attraction images were stored in the /assets/images/ directory.

Icons, logos, and background images were optimized for fast loading.

Default placeholders were used for testing missing media cases.

2. Security and Privacy Measures

- ➢ Sensitive fields (e.g., passwords) were hashed using secure algorithms (e.g., bcrypt).
- ➢ User data was anonymized for testing where necessary to protect privacy.
- ➢ Input validation and sanitation were applied to prevent SQL injection and XSS attacks.

3. Data Backup

- Regular backups of the database were maintained during development to prevent loss.
- A versioned backup system ensured changes could be rolled back during testing or system failure.

4. Testing Scenarios

Prepared datasets were used in test scenarios such as:

- Tourist registering and booking tickets.
- System administrator managing ticket inventory.
- Payment processing and confirmation.
- Report generation and analytics.

4.3. Implementation

The development was carried out in modules to ensure clarity and ease of debugging:

4.3.1. Tourist Module

Tourists can view packages

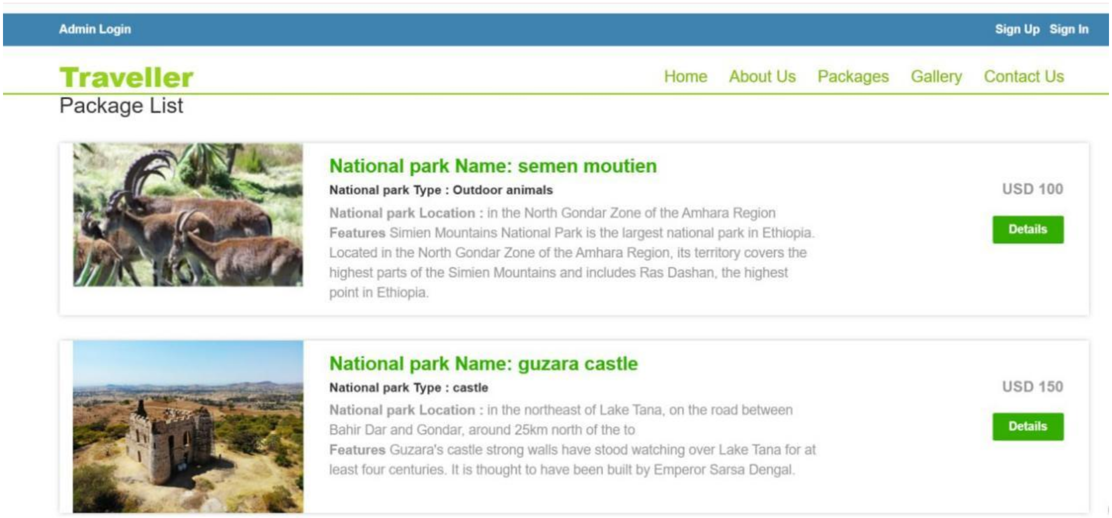

Figure 4.1 packages

Tourists can view details

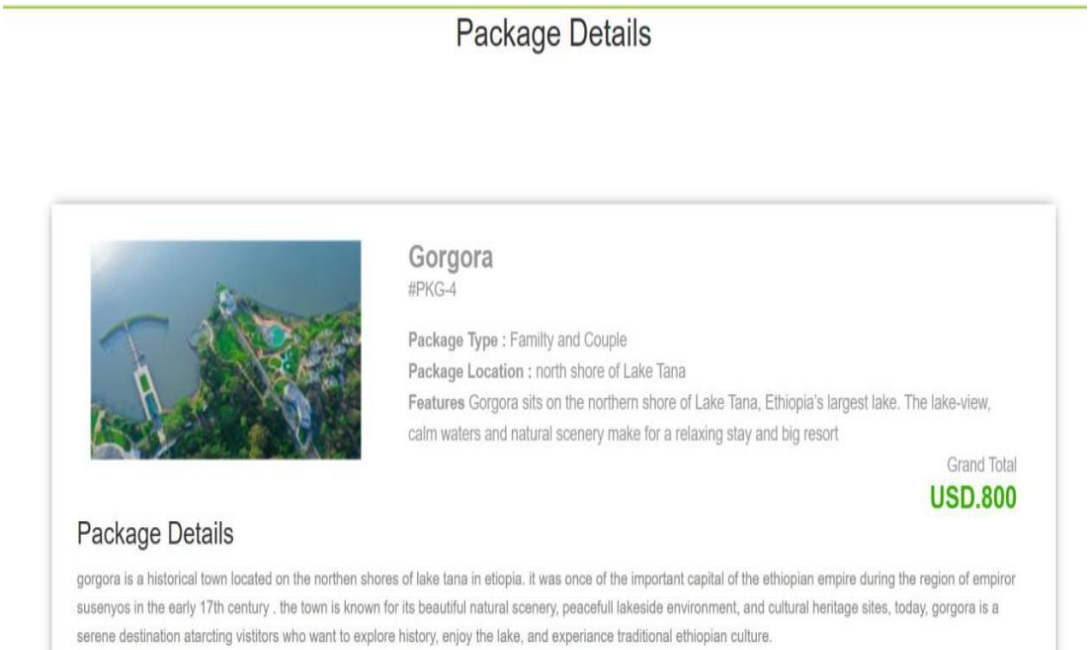

Figure 4.2 Package details

Tourists can register

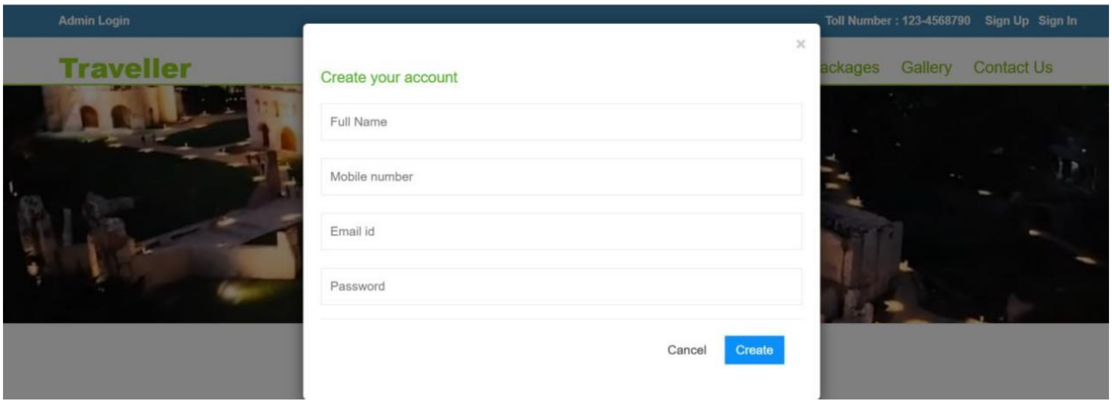

Figure 4.3 Registration page

Tourists can book tickets

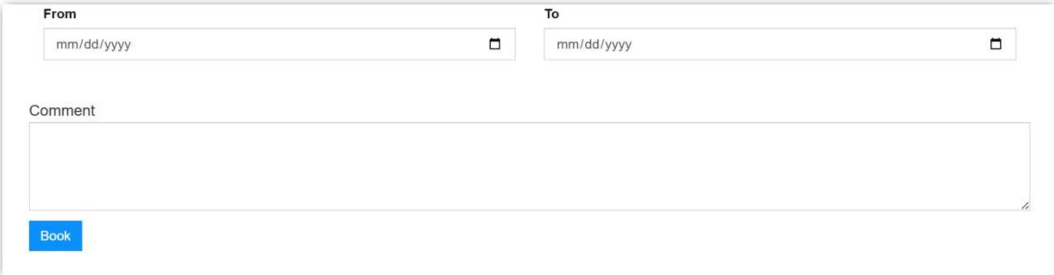

Figure 4.4 Book page

Tourist can edit profile.

Figure 4.5 Profile edit page

Tourists can view booking processes

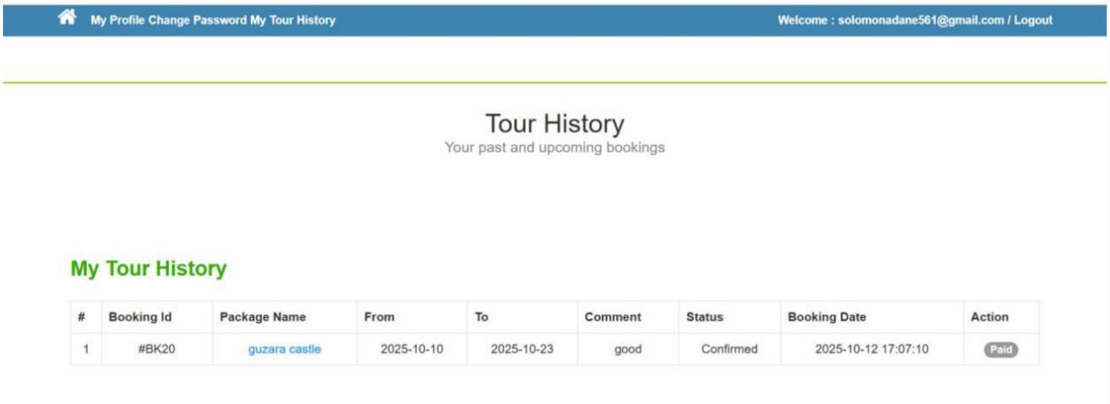

Figure 4.6 Registration page

4.3.2. System Admin Module

Admin can manage tickets and confirm booking

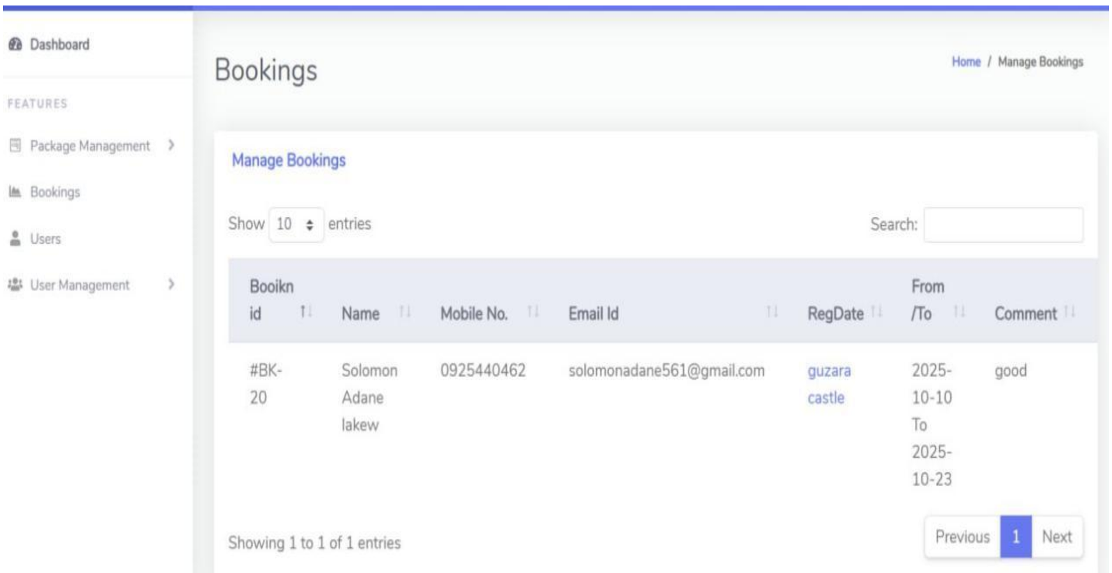

Figure 4.3 Booking confirmation page

Admin can add another packages

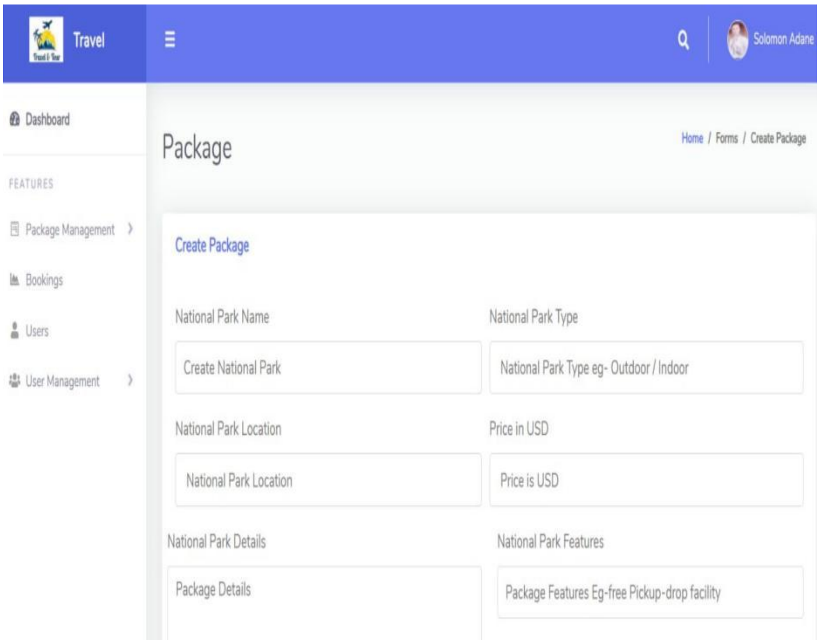

Figure 4.8 Package adding page

Payment Integrated with Chapa for seamless and secure payment processing. Ensures real-time transaction verification for fast user confirmation. Supports instant SMS alerts to validate each payment step. Delivers a smooth, reliable checkout experience for every user

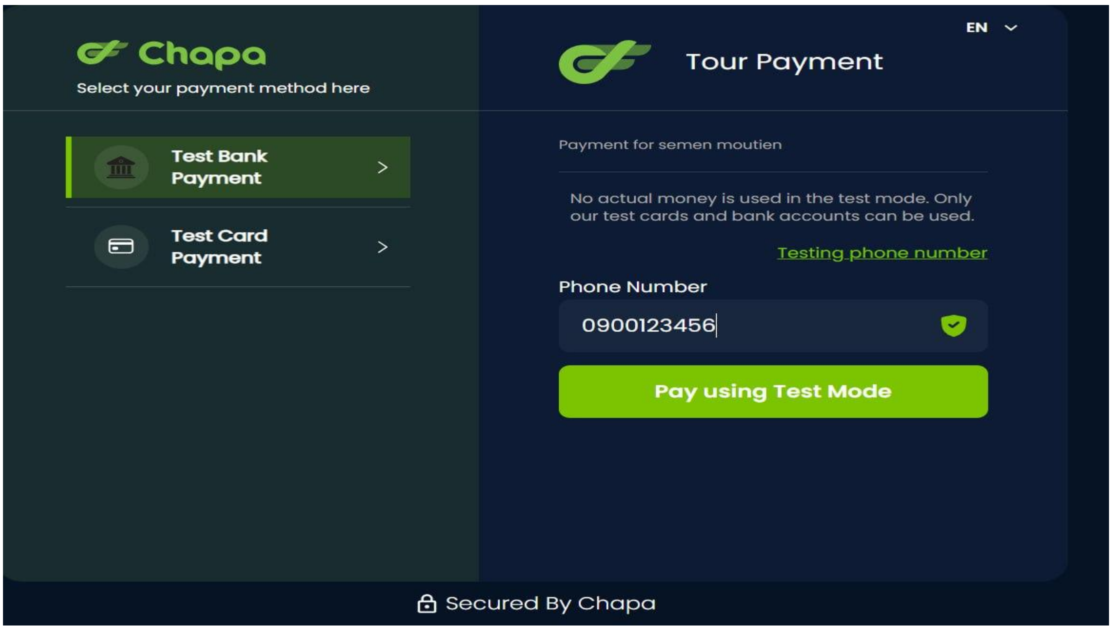

Figure 4.9 Payment method with chapa


Chapa

RECEIPT

Receipt From

Test Business

TIN Test TIN

Phone No. 0900123456

Address Test Address

Chapa Financial Technologies S.C

TIN 0071406415

VAT Reg. 18595770010

Phone No. +251-960724272

Website chapa.co

PAYMENT DETAILS	RCAP03PKMgoCb5E
Payer Name / የከፋይ ስም	**semen moutien Booking-40**
Email Address / ኢሜል አድራሻ	**solomonadane561@gmail.com**
Payment Method / የክፍያ መንገድ	**Test**
Status / ሁኔታ	**Paid / ተከፍሏል**
Payment Date / የክፍያ ቀን	**09-12-2025**

Figure 4.10 Receipt with chapa

CHAPTER FIVE

Result and discussion

The implementation of the Tour Ticket Reservation System produced strong and promising results, successfully achieving its primary objectives of modernizing the ticketing process, improving service efficiency, and enhancing communication between tourists and system administrators. The system efficiently handled core functionalities such as user registration, attraction browsing, ticket booking, payment processing, and automated confirmation notifications. During functional testing, each module performed accurately, showing reliable data handling and smooth workflow transitions across the booking and payment processes. The user interface was intuitive, clean, and mobile-responsive, allowing tourists with different technical backgrounds to easily navigate the platform and complete ticket reservations. Furthermore, role-based access control and security mechanisms were effectively integrated, protecting user information and ensuring only authorized personnel could manage system data. Although minor issues were initially encountered during real-time inventory updates and payment gateway integration, these were resolved through iterative testing. Overall, the system demonstrated good performance, usability, and scalability, making it a valuable digital solution for improving ticketing operations, reducing manual workload, and enhancing the overall tourism experience in Gondar.

CHAPTER SIX

Conclusion and recommendation

Conclusion

The Online Tour Ticket Reservation System for Gondar successfully modernizes ticket purchasing operations by replacing traditional manual processes with an efficient, automated, and user-friendly digital platform. The system enhances coordination between tourists and administrators by providing centralized access to attraction information, real-time ticket availability, secure online payments, and instant booking confirmations. Through its modular architecture, role-based access control, and responsive interface, the system ensures data accuracy, operational reliability, and improved service delivery. These advancements contribute to a more organized tourism workflow and elevate the overall visitor experience, positioning Gondar as a more accessible and competitive destination for both local and international tourists.

Recommendations

Future advancements of the system should include QR-code ticket scanning at attraction entrances, an online refund module, and smart scheduling tools to better control peak tourist traffic. Security can be strengthened by integrating two-factor authentication, encrypting sensitive data, and implementing automated security alerts. Developing a dedicated Android/iOS mobile application would provide tourists with quicker access to booking services and real-time updates. A multi-language interface—featuring Amharic, English, French, and other global languages—would ensure accessibility for international visitors. Additionally, incorporating AI capabilities can help analyze user behavior, predict demand peaks, and enable dynamic pricing for improved operational efficiency. The system should also support a Multi-Banking feature to allow secure payments across various local and international banks, reducing dependency on a single provider. Furthermore, Multi-Currency Support should be included to let users pay in their preferred currency with real-time conversion, ensuring clarity, convenience, and accuracy for global tourists.

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